

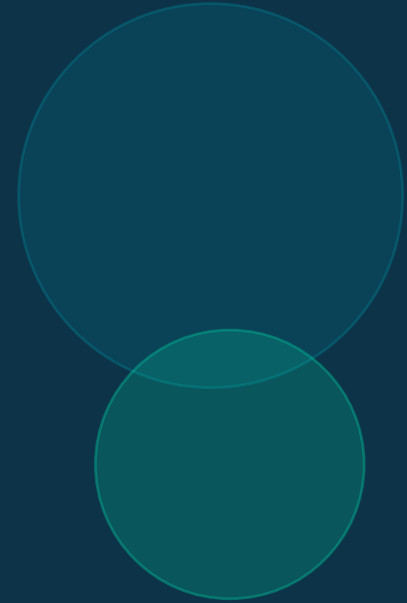
Forecasting Monthly Maize Prices Across Kenyan Counties

To Support Food Security Planning

Sharon Rosina Wamalwa

University of South Dakota | April 2026

Instructor: Thomas Tiahrt, Ph.D.



Introduction & Problem

Why Maize Forecasting Matters

- Primary staple food in Kenya — consumed across all income groups
- Price volatility directly causes acute food insecurity
- Governments historically react to crises after they occur
- No reliable forward-looking forecasting tools exist at county level

Study Objective

- Develop & evaluate 4 time-series forecasting models for monthly maize prices across 5 Kenyan counties
- Generate 24-month future price projections to support proactive food security planning
- Data: WFP Kenya Food Prices dataset — January 2006 to 2025

Research Question: Which forecasting model — Naïve, ETS, ARIMA, or TSLM — produces the most accurate 24-month forecasts for each county, and what do the results imply for food security?

Agricultural

Uasin Gishu

Arid / ASAL

Turkana • Marsabit

Urban

Nairobi • Mombasa

Data Acquisition & Preparation

Data Source: World Food Programme (WFP) Kenya Food Prices — Humanitarian Data Exchange (HDX)

Coverage

28 Kenyan counties
Jan 2006 – Nov 2025
Monthly commodity prices

Commodity

'Maize (white)' 2006–2021
and 'Maize (white, dry)'
2021–2025 — combined

Unit Standardization

Marsabit & Turkana: KES/KG
Others: KES per 90KG sack
All converted to KES/KG

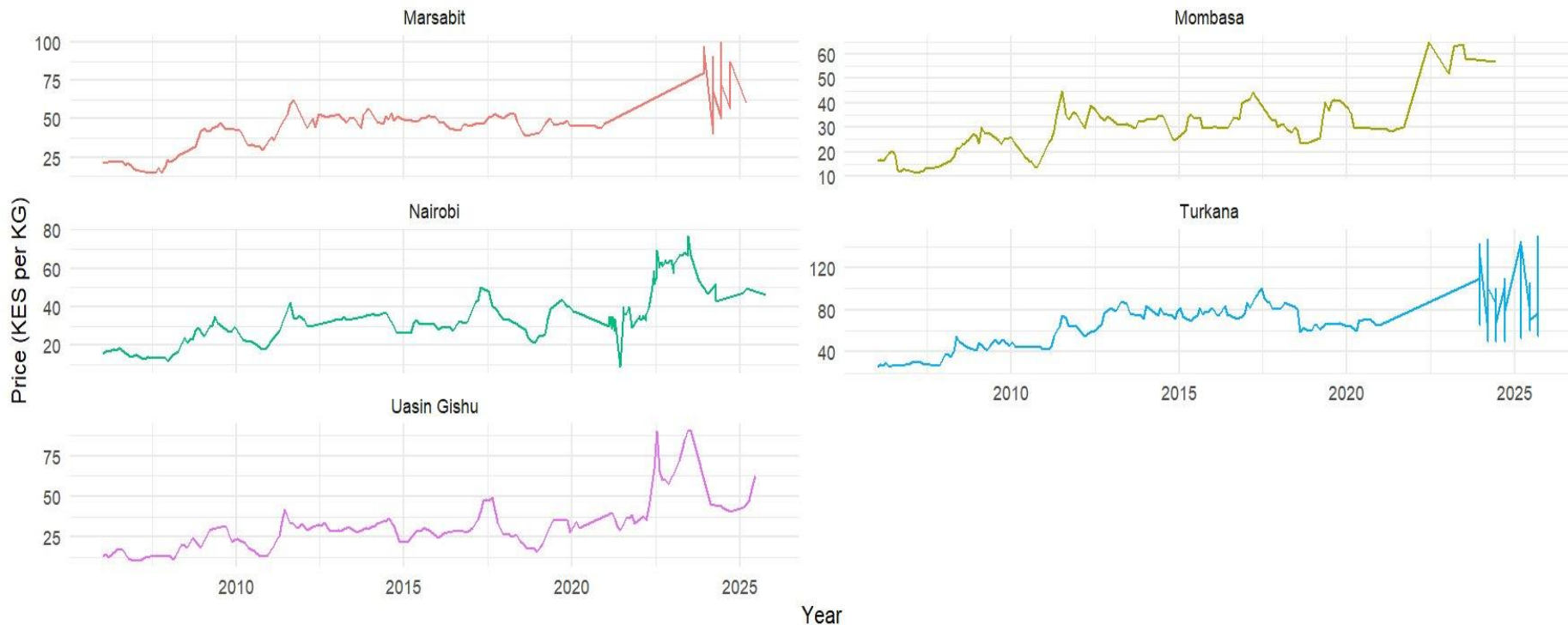
Nakuru Removed

Data started 2015 only
11-month gap in 2020
Too irregular for modeling

Final Dataset: 5 Counties | Marsabit • Mombasa • Nairobi • Turkana • Uasin Gishu | Turkana: 267 observations (most) | Mombasa: 125 (fewest) | All exceed minimum 48 obs. required

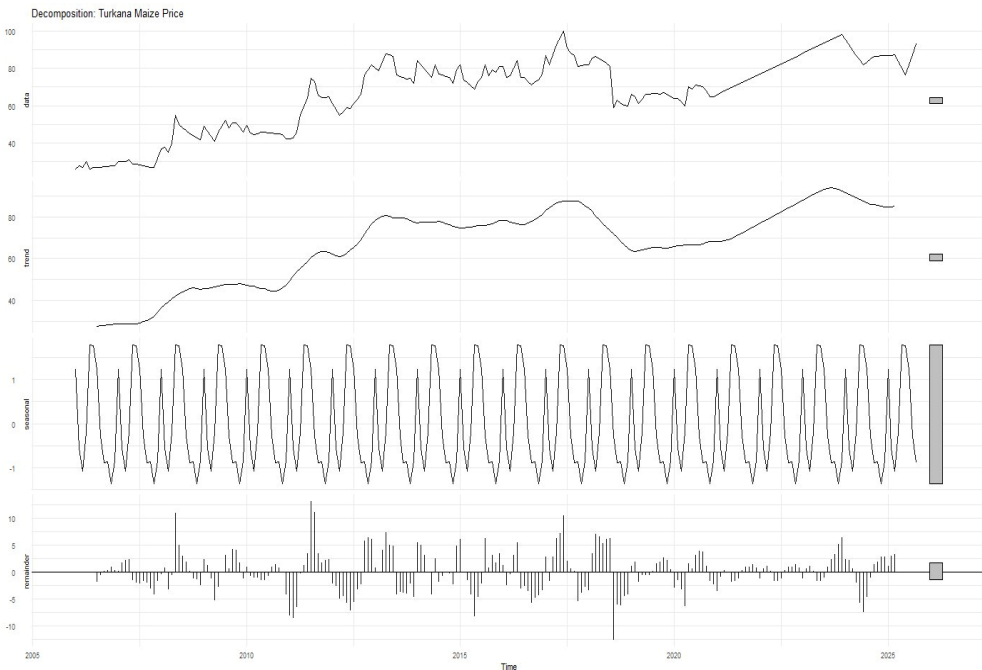
Maize Price Trends — All 5 Counties (2006–2025)

Maize Price Per KG by County (2006–2025)



All counties show strong upward trends with sharp spikes in 2022–2024 driven by drought, COVID supply disruptions, and the Russia-Ukraine conflict.

Data Exploration — Key Findings



Decomposition: Turkana — clear trend, small seasonal, irregular spikes

4 Consistent Findings

1

Strong upward trend

All 5 counties show clear long-run price increases 2006–2025

2

Weak seasonality

Seasonal patterns present but small vs. trend and shocks

3

Non-stationary series

ACF slowly decays, single PACF spike at lag 1 — unit root confirmed

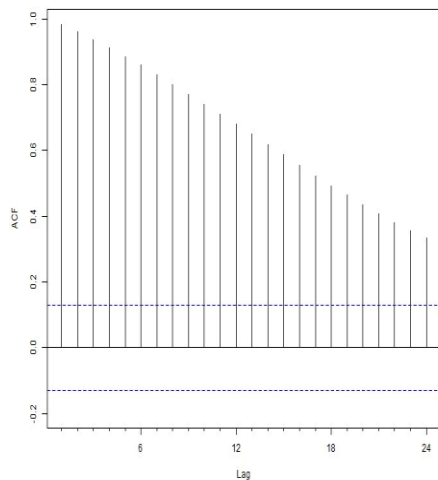
4

Large irregular shocks

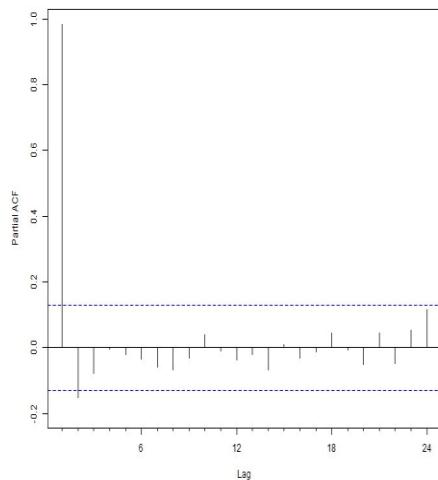
Major spikes 2009–2011 and 2021–2023 from droughts & global shocks

ACF & PACF Analysis — Confirming Non-Stationarity

ACF: Marsabit Maize Price

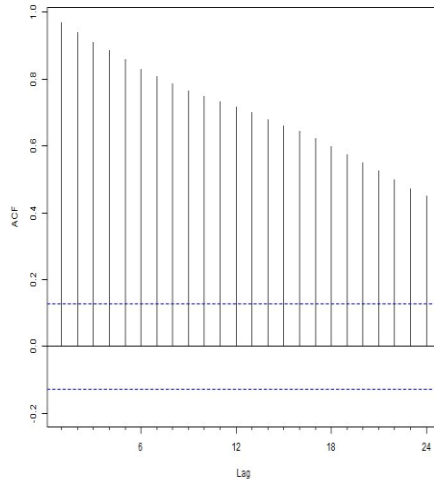


PACF: Marsabit Maize Price

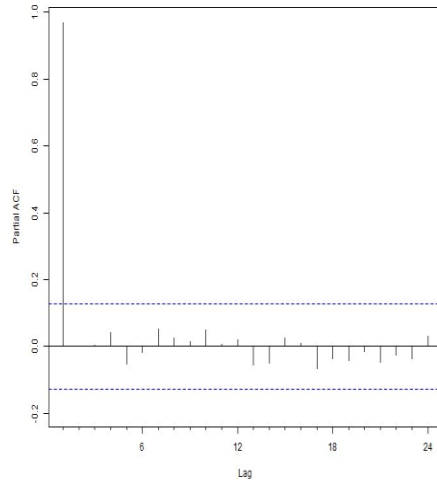


Marsabit ACF/PACF

ACF: Turkana Maize Price



PACF: Turkana Maize Price



Turkana ACF/PACF — most persistent of all 5 counties

Pattern consistent across ALL 5 counties: Slowly decaying ACF + single dominant PACF spike at lag 1

- Classic hallmark of a non-stationary integrated process (unit root present)
- First differencing required before fitting ARIMA models | Formally confirmed by Augmented Dickey-Fuller (ADF) test

Data Transformation

1

Missing Month Interpolation

All counties had gaps from irregular data collection. Linear interpolation filled missing months by drawing a straight line between nearest known values. Turkana was the only county with zero missing months (267 complete observations).

2

Train / Validation Split

Last 24 months held back as validation set — models never saw this data during training. All models fitted exclusively on training data and evaluated on validation set to ensure honest out-of-sample accuracy measurement and avoid overfitting.

3

Predictor Variable Creation (for TSLM)

Three new variables created: (a) Time index — sequential counter capturing the upward trend. (b) Month variable — 12 calendar month categories capturing seasonality. (c) Lag variable — previous month's price capturing short-term price persistence.

Four Forecasting Models — Motivation & Overfitting Avoidance

Naïve

Baseline benchmark. Forecasts = last observed price. Any model failing to beat it has no practical value.

ETS

Error, Trend, Seasonality. Assigns exponentially decreasing weights to past data — recent shocks matter more. Auto-selects best specification via AICc.

ARIMA

Autoregressive Integrated Moving Average. Handles non-stationarity via differencing. Auto-selects optimal (p,d,q) parameters using AIC.

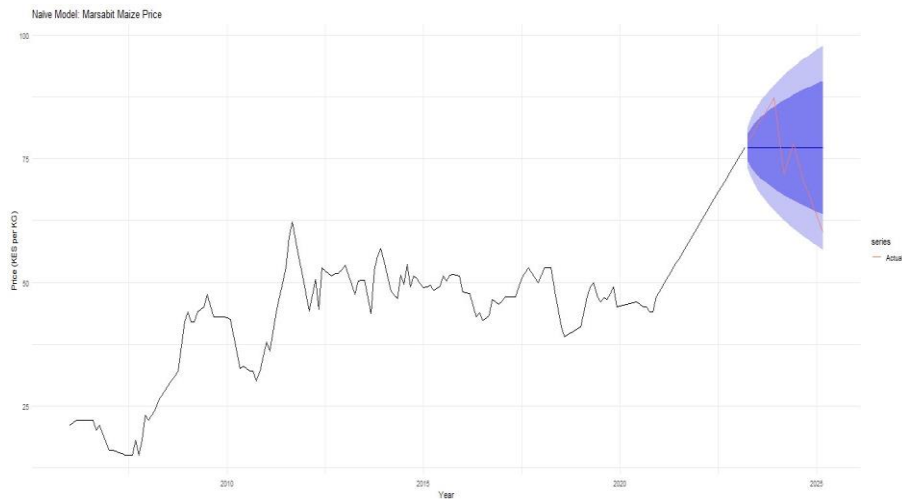
TSLM

Time Series Linear Model. Regresses prices on time trend + monthly seasonal dummies. Interpretable coefficients — ideal for policy communication.

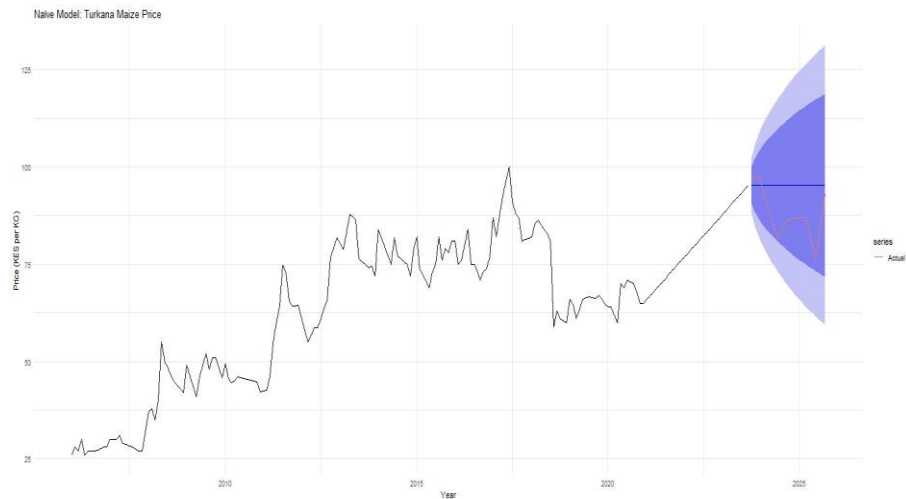
Overfitting Avoidance

- ① All models trained on training set only, tested on held-out 24-month validation set
- ② AIC & BIC penalize model complexity — favoring parsimonious specifications
- ③ Naïve model retained as benchmark — guards against unnecessarily complex models
- ④ 24-month validation horizon — long enough to expose overfitting not visible in shorter windows

Naïve Model Results — Baseline Benchmark



Naïve Model — Marsabit (MASE: 0.845 ✓)

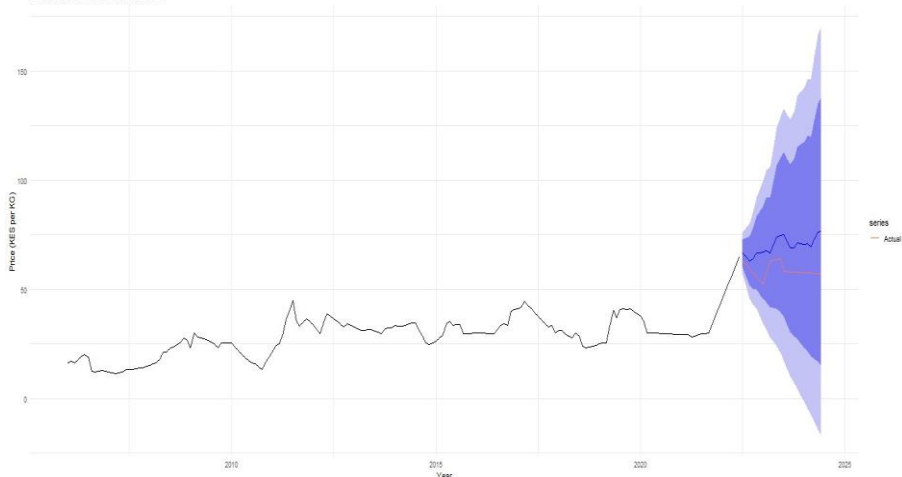


Naïve Model — Turkana (MASE: 0.936 ✓)

County	Validation MASE	Beat Benchmark?	Winner?
Marsabit	0.845	✓ Yes	✓ Naïve Wins
Mombasa	0.844	✓ Yes	✓ Naïve Wins
Nairobi	0.641	✓ Yes	✗ TSLM beats it
Turkana	0.936	✓ Yes	✗ TSLM beats it
Uasin Gishu	4.274	✗ No	✓ Still wins (all others worse!)

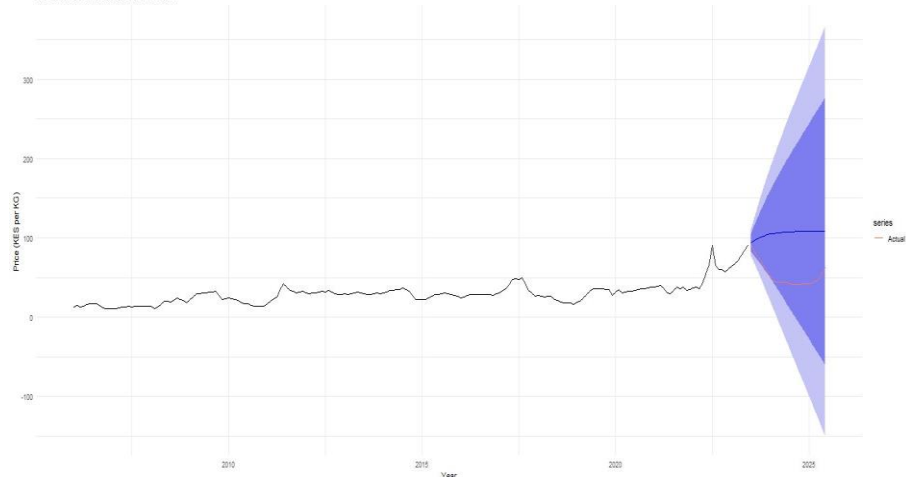
ETS Model Results — Exponential Smoothing

ETS Model: Mombasa Maize Price



ETS Mombasa — MASE: 1.509 (worse than Naïve). Unrealistic forecast above KES 150/kg

ETS Model: Uasin Gishu Maize Price



ETS Uasin Gishu — MASE: 5.982 (worst result). Intervals extend below zero — impossible!

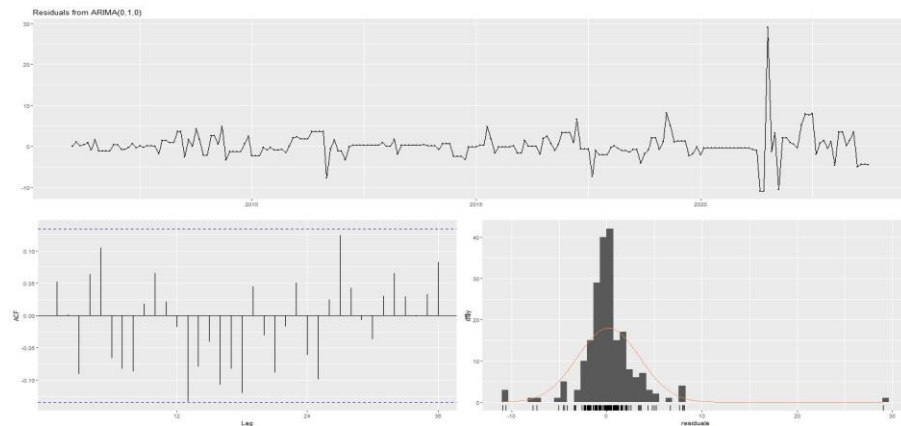
ETS Overall Verdict

ETS did NOT win any county. Nairobi & Turkana received ETS(A,N,N) — effectively identical to the Naïve model. Mombasa received complex ETS(M,Ad,M) where multiplicative components amplified forecast errors unrealistically. Uasin Gishu's 2022 price spike misled the model into projecting unrealistically high future prices (MASE: 5.982, MAPE: 113.9%). Overall worst performing model family.

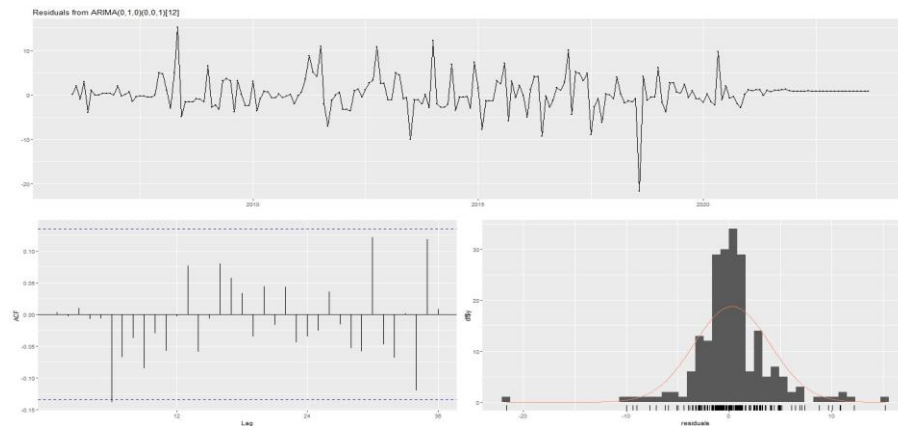
ARIMA Model Results

ARIMA Specifications Selected by auto.arima()

County	Model	Validation MASE	Beat Naïve?
Marsabit	ARIMA(0,1,2) with drift	1.125	✗ No
Mombasa	ARIMA(1,1,0)	1.085	✗ No
Nairobi	ARIMA(0,1,0) — pure random walk	0.641	= Same as Naïve
Turkana	ARIMA(0,1,0)(0,0,1)[12] — seasonal	1.031	✗ Marginally worse
Uasin Gishu	ARIMA(0,1,1)	4.382	✗ Much worse



Nairobi ARIMA(0,1,0) residuals — clean white noise. Ljung-Box $p=0.1858$ ✓



Turkana ARIMA seasonal residuals — Ljung-Box $p=0.8506$ ✓ (strongest result)

TSLM Results — Best Model: Nairobi

Nairobi MASE

0.241

Best result in entire study

Nairobi MAPE

4.37%

Only 4.37% avg error

Nairobi RMSE

KES 2.62

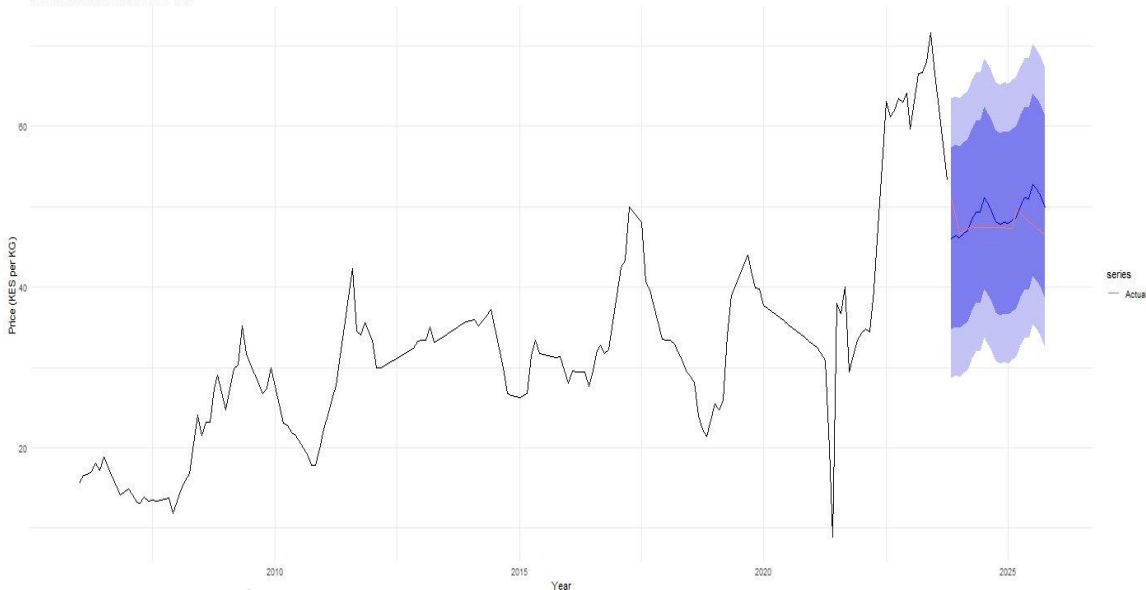
Per kg avg forecast error

Turkana MASE

0.913

TSLM wins for Turkana too

TSLM Model: Nairobi Maize Price



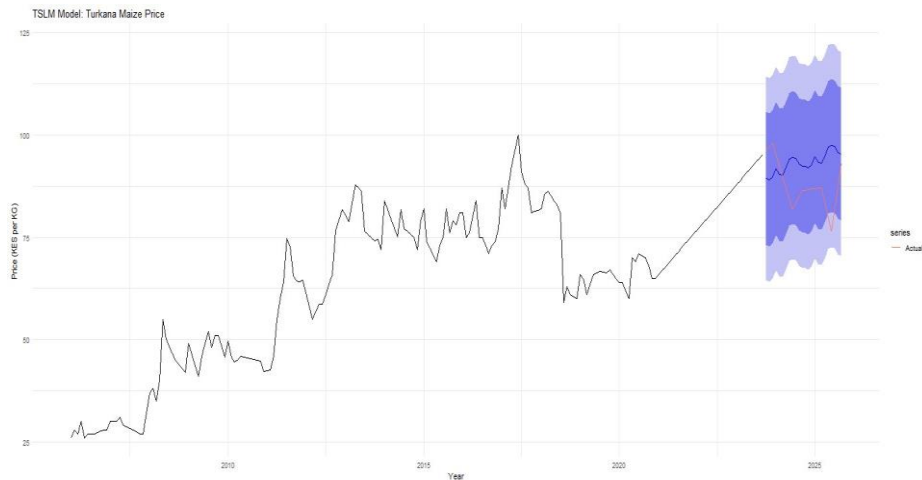
TSLM Nairobi — forecast line tracks actual prices remarkably closely throughout all 24 validation months

Why TSLM Won for Nairobi

- Nairobi prices followed a clear, predictable linear trajectory during the validation period
- Long-run upward trend continued steadily without dramatic unexpected shocks
- Seasonal structure was consistent with historical patterns
- TSLM explicitly models trend + seasonal dummies — perfect fit for this behavior

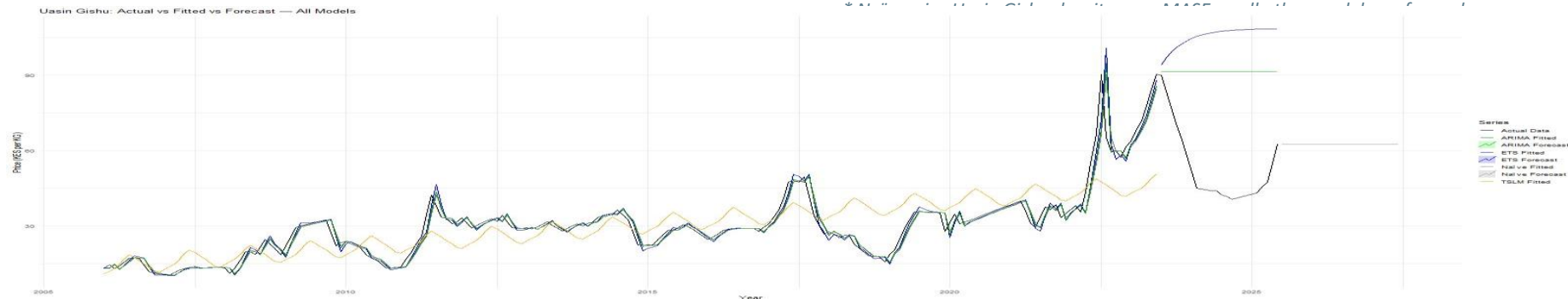
Also winner for Turkana
(MASE: 0.913 — strong persistent upward trend)

Model Comparison — Validation MASE Across All Models & Counties



TSLM Turkana — upward trend projection, MASE: 0.913 ✓

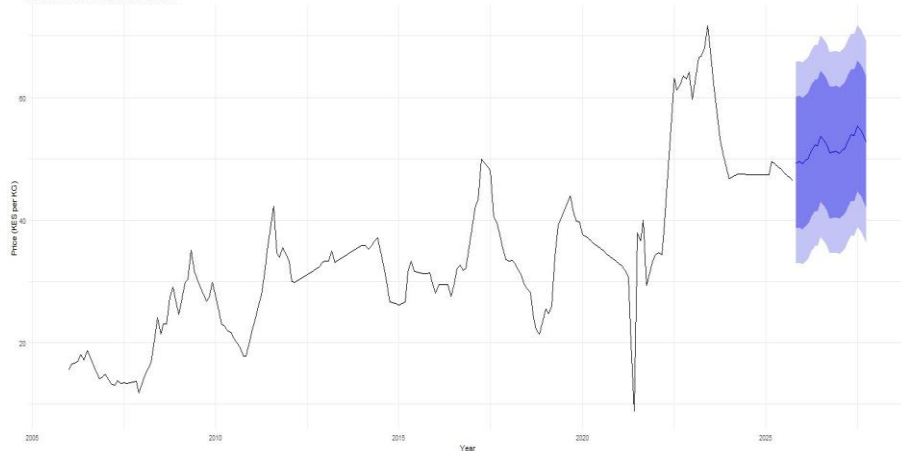
County	Naïve	ETS	ARIMA	TSLM	Winner
Marsabit	0.845 ✓	0.929	1.125	1.662	Naïve
Mombasa	0.844 ✓	1.509	1.085	2.355	Naïve
Nairobi	0.641	0.641	0.641	0.241 ★	TSLM
Turkana	0.936	0.936	1.031	0.913 ✓	TSLM
Uasin Gishu	4.274*	5.982	4.382	1.225	Naïve*



Uasin Gishu — all 4 models vs actual. ETS forecast (blue curve) wildly overestimated prices. None captured the 2022 spike reversal.

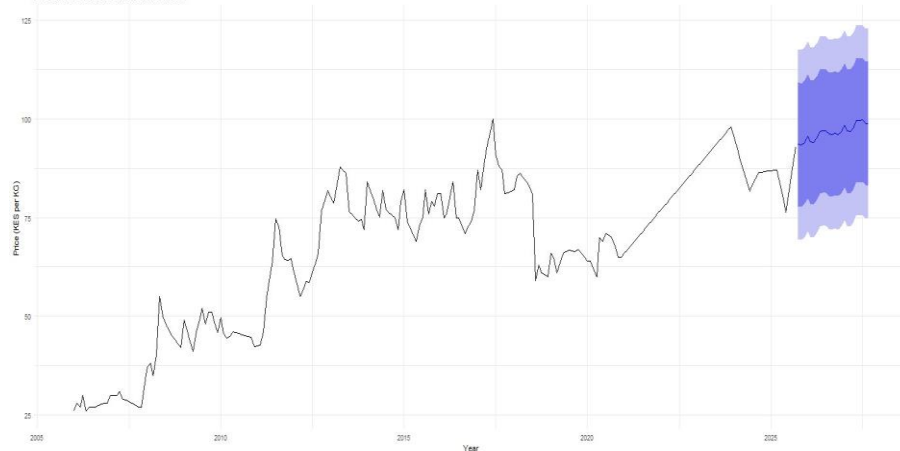
24-Month Future Forecasts — Nov 2025 to Oct 2027

24-Month Forecast: Nairobi Maize Price



Nairobi (TSLM) — Most reliable forecast. Stable KES 49–55/kg with mild seasonality

24-Month Forecast: Turkana Maize Price



Turkana (TSLM) — ALARM: Rising toward KES 99/kg by Oct 2027

County	Model Used	Forecast	Policy Message
Marsabit	Naïve	KES 60.00/kg flat	Monitor — vulnerable to shocks
Mombasa	Naïve	KES 57.00/kg flat	Relatively stable — port access helps
Nairobi	TSLM	KES 49–55/kg seasonal	Most reliable forecast — stable outlook
Turkana	TSLM	KES 93-99/kg rising	URGENT — pre-position food reserves!
Uasin Gishu	Naïve	KES 62.50/kg flat	High uncertainty — monitor harvest & inputs

Conclusions & Future Research

Key Findings

- No single model dominated all counties — price dynamics differ by county type
- TSLM won for Nairobi (MASE 0.241 — best result in entire study) and Turkana (0.913)
- Naïve model won for Marsabit, Mombasa & Uasin Gishu — confirmed random walk behavior
- ETS performed worst — failed to beat Naïve in any county
- ARIMA comparable to Naïve — Nairobi's ARIMA(0,1,0) is literally a random walk

Food Security Alarm: Turkana prices projected to approach KES 100/kg by 2027 — Kenya's most vulnerable county. Policymakers must act now: pre-position grain reserves, strengthen road infrastructure, target food assistance.

Limitations

- Univariate models only — no rainfall, fuel prices, or exchange rates
- Linear interpolation introduced artificial smoothness in 4 of 5 counties
- 2021–2025 extraordinary volatility may limit historical pattern reliability

Future Research Directions

- Multivariate models: add rainfall, satellite vegetation indices, fuel prices & exchange rates
- Machine learning: Gradient Boosting & LSTM neural networks for non-linear pattern capture
- Extend to all 28 Kenyan counties in the WFP dataset for comprehensive national coverage